HW#5 Domain-Specific Accelerator

Chun-Jen Tsai
NYCU
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Homework Goal

- In this homework, you must integrate a domain-specific accelerator (DSA) in the Aquila SoC to improve the inferencing speed of a CNN neural network

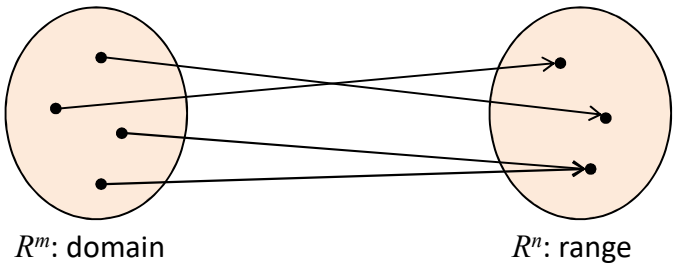
- Your tasks:
  - Design an AI accelerator HW IP
  - You can use FP32, FP16, or even INT8 formats in your IP
  - If you plan to use FP32/16, use Xilinx's floating-point IP
  - If you use INT8 format, the accuracy must be maintained

- You should upload report & code to E3 by 12/22, 23:55.

Neural Network as Computers

- All computing systems are used to compute functions:

$$f: R^m \rightarrow R^n$$

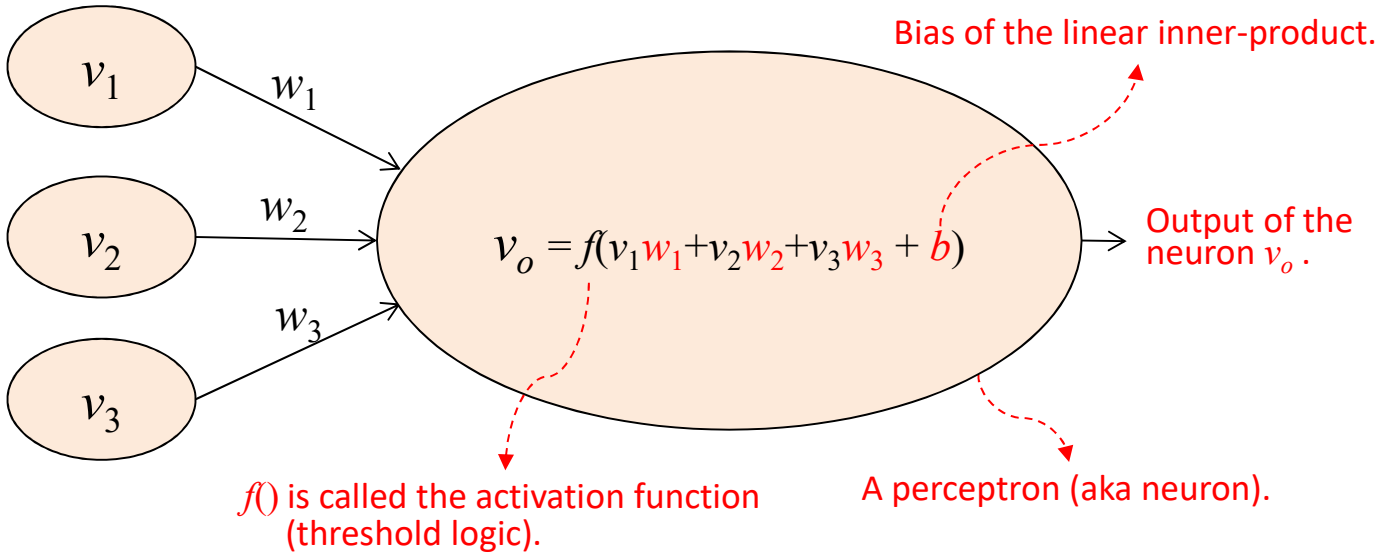


- In 1957, Kolmogorov proved that any continuous function can be decomposed into a linear combination of $(2m+1)+n$ different $R^m \rightarrow R$ linear functions

† A. N. Kolmogorov, "On the representation of continuous function of many variables by superpositions of continuous functions of one variable and addition," *Doklady Akademii Nauk USSR*, 114(5):953-956, 1957.

Basic Neural Network Components

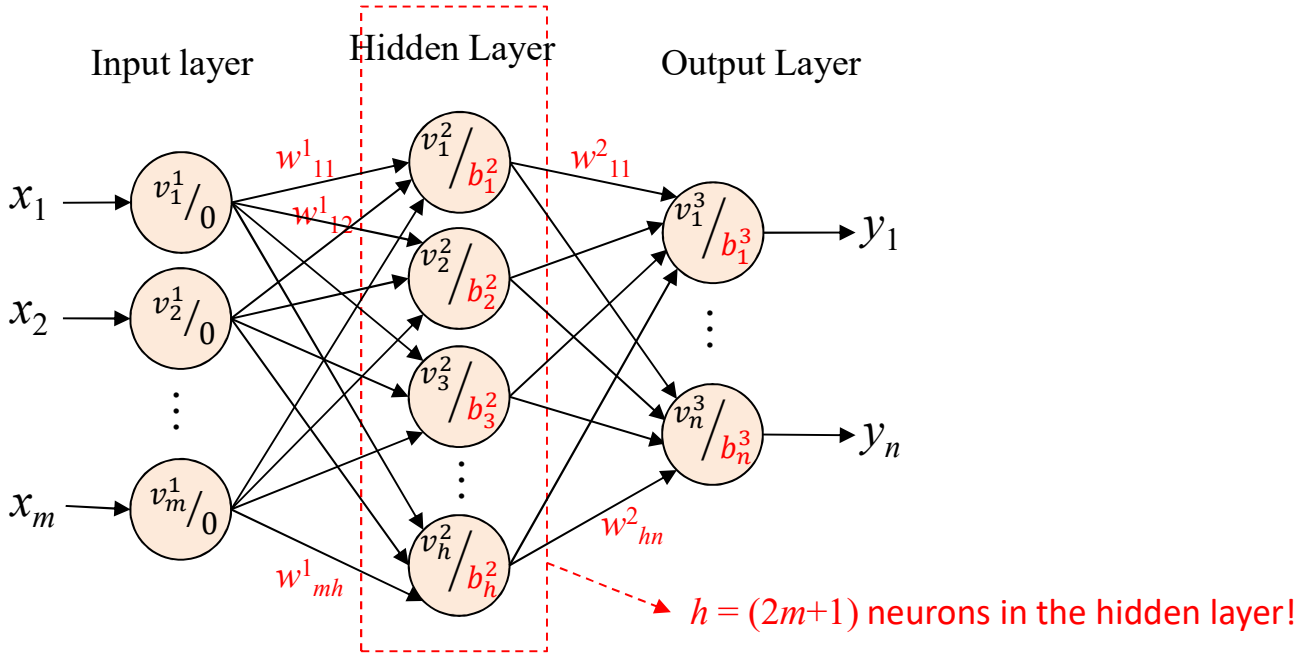
- W. McCulloch and W. Pitts proposed a computing element, threshold logic, for Artificial Neural Network (ANN) in 1943:



† W. McCulloch and W. Pitts, "A Logical Calculus of Ideas Immanent in Nervous Activity". Bulletin of Mathematical Biophysics. **5** (4), 1943, pp. 115–133.

A Universal Neural Network

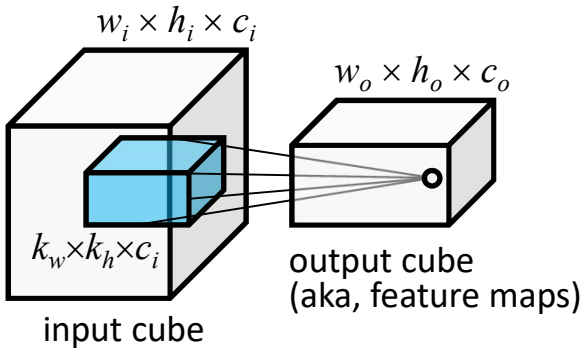
- By Kolmogorov, a continuous function f: $R^m \rightarrow R^n$ can be computed using 3 layers of perceptrons



  - The "program" of a neural network is in the weights, $\{w_{ji}^{\ k}, b_i^{\ k}\}$
  - This architecture is called Multi-Level Perceptrons (MLP)

Different Neural Network Layer Types

- Convolutional Layer
  - Computes 3D dot product btw. the input data & the kernel
  - Each neuron in the output cube of the same channel shares the same kernel

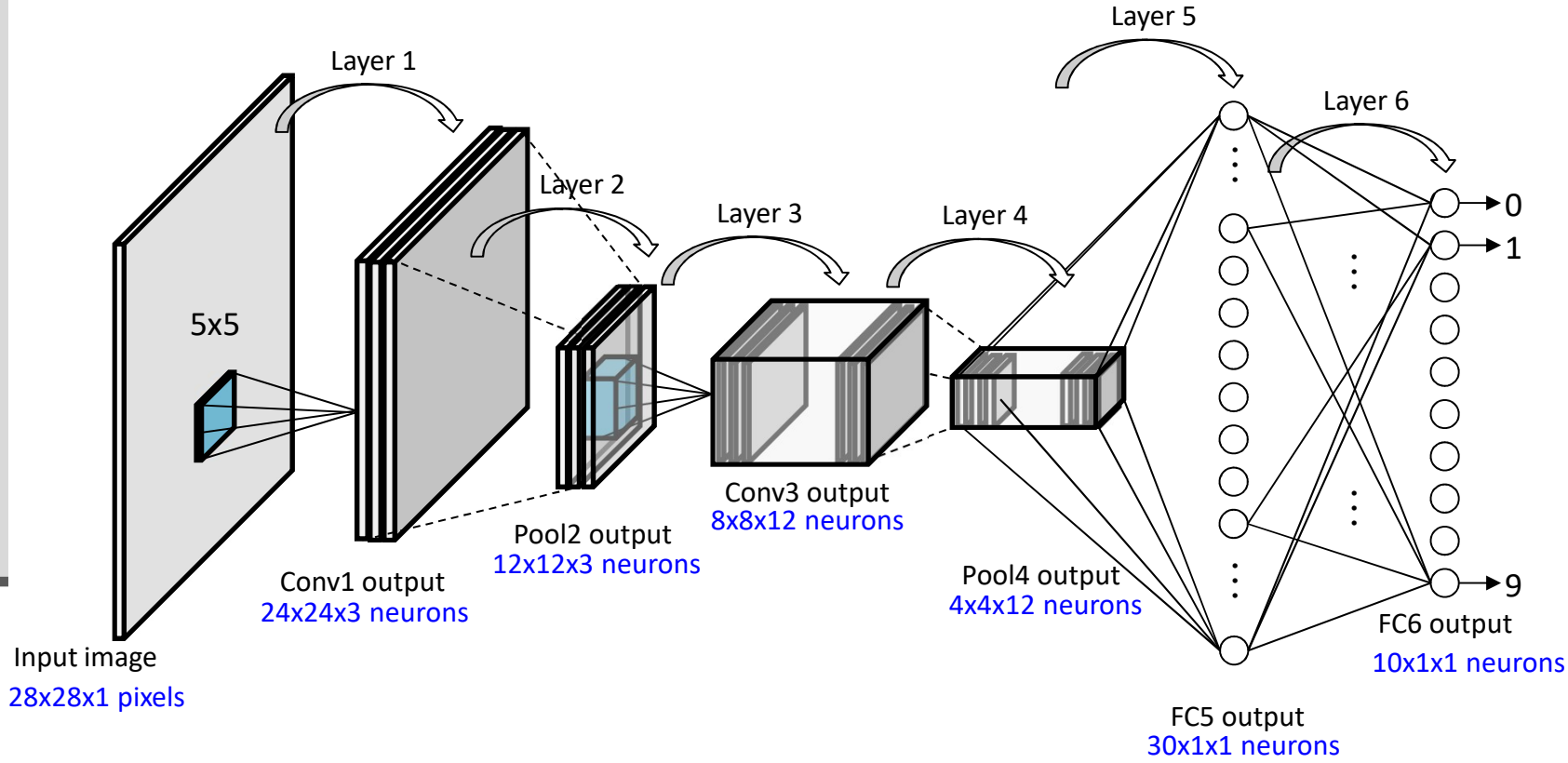


- Avg-Pooling Layer
  - Reduce the width & height of the input feature maps by averaging the neuron values in a small $m \times n$ rectangles of the same channels

- Fully-Connected Layer
  - As in the MLP

Convolutional Neural Network (CNN)

- The CNN model used in this HW:



Hand-Written Character Recognition

- In this HW, we use a 6-layer CNN for hand-written character recognition
- The neural network weights were trained using the MNIST dataset of 60,000 images:
  - Each image has 28×28 pixels

The HW/SW for HW#5

- The C code for HW#5 is in the directory `cnn_ocr`
- The source tree of the `cnn_ocr` program:

```
cnn_ocr2/
├── Makefile
├── data/
│   ├── test-images.dat
│   ├── test-labels.dat
│   └── weights.dat
│
├── inc_cnn/*.h
│
├── file_read.c
├── file_read.h
├── cnn_ocr.c
└── cnn_ocr.ld
```

Data files to be copied to the SD card.

The header files of the CNN model

The source code of the OCR application

  - Note that the low-level SD I/O routines are in `elibc/fileio/`

Building an FPU-enabled Aquila

- To build the `cnn_ocr` program with or without FPU, you have to change the top two lines of `Makefile`:
  - With FPU

```
# ==========================================================================
#  This Makefile builds an application *.elf for Aquila.
#
#  Chun-Jen Tsai, Aug/28/2025
# ==========================================================================
RISCV_ABI = -march=rv32imad_zicsr_zifencei -mabi=ilp32d
GCC_LIB = rv32imad/ilp32d

CROSS = riscv32-unknown-elf
CCPATH = $(RISCV)/bin
```

  - Without FPU

```
# ==========================================================================
#  This Makefile builds an application *.elf for Aquila.
#
#  Chun-Jen Tsai, Aug/28/2025
# ==========================================================================
RISCV_ABI = -march=rv32ima_zicsr_zifencei -mabi=ilp32
GCC_LIB = rv32im/ilp32

CROSS = riscv32-unknown-elf
CCPATH = $(RISCV)/bin
```

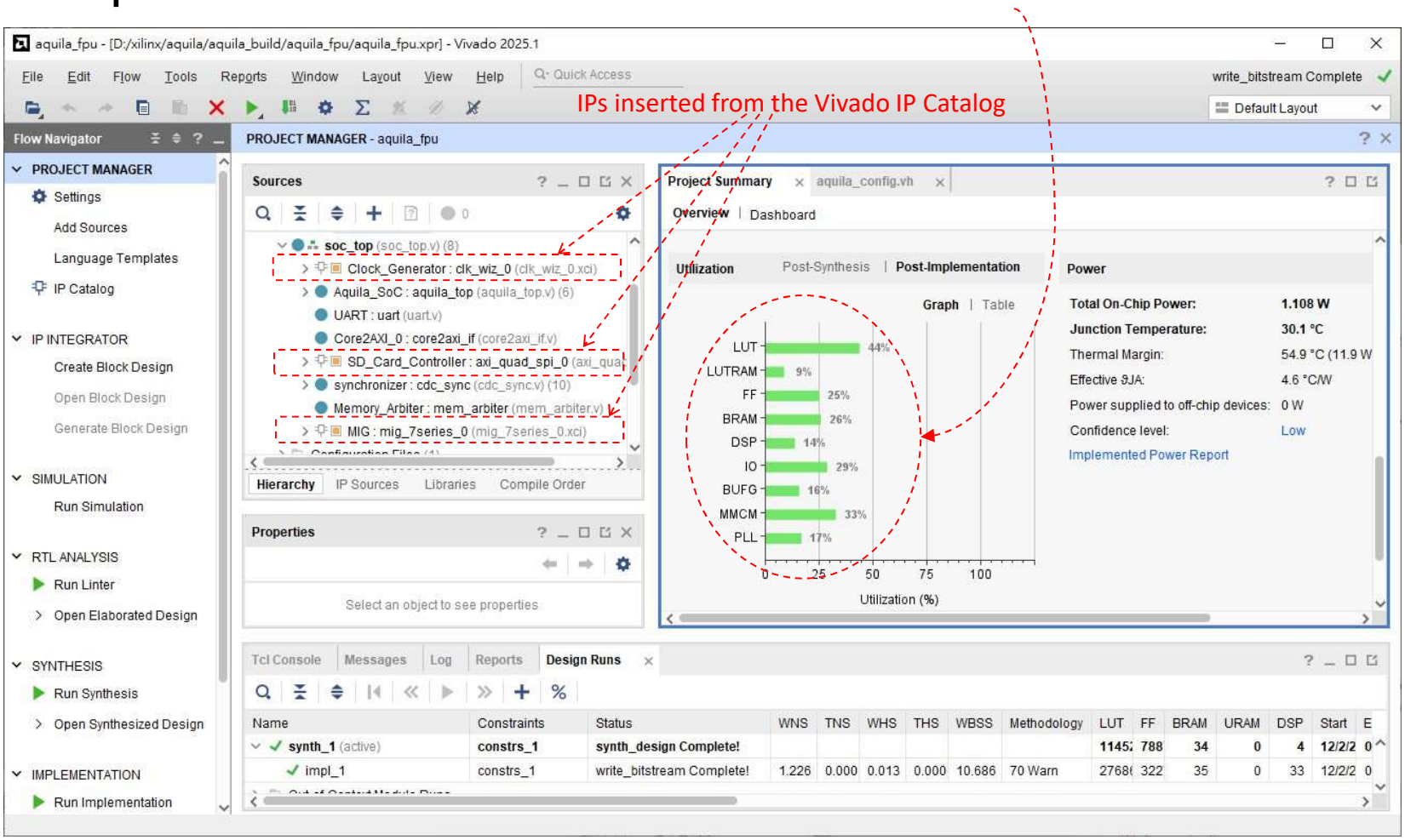
Note: don't type 'rv32ima' here!

Important Notice

- When you Integrate your AI Accelerator with Aquila, you shall use the Aquila without FPU!
- But we will compare the performance of your system against Aquila with FPU
- To beat Aquila with FPU, it would probably be easier to simply use INT8 for your AI model, but you have to perform proper scaling to avoid dropping accuracy
- To encourage you to use FP IP, you get extra points for doing this

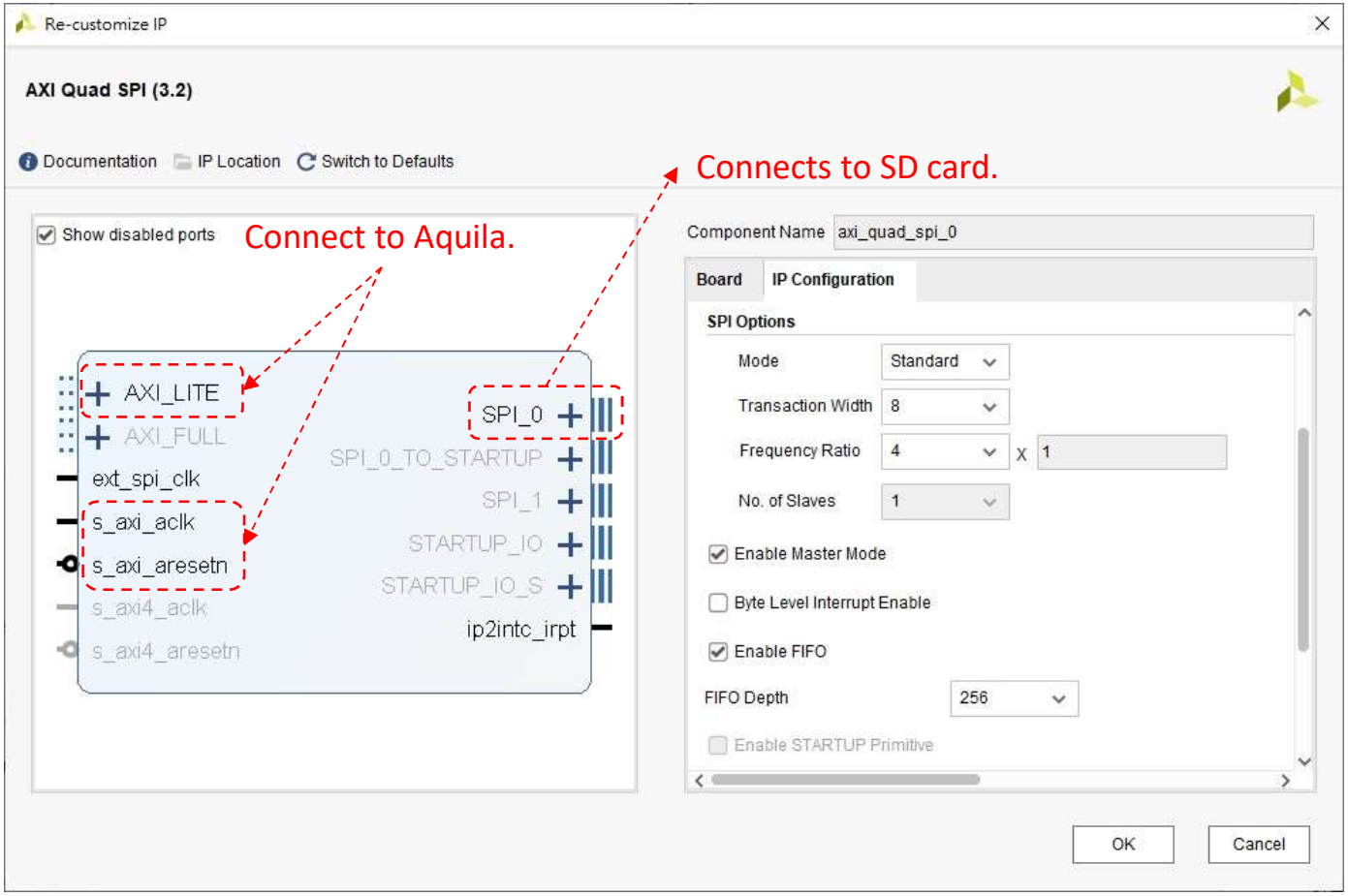
The HW Workspace for HW#5

- Aquila with FPU used a lot of resources:



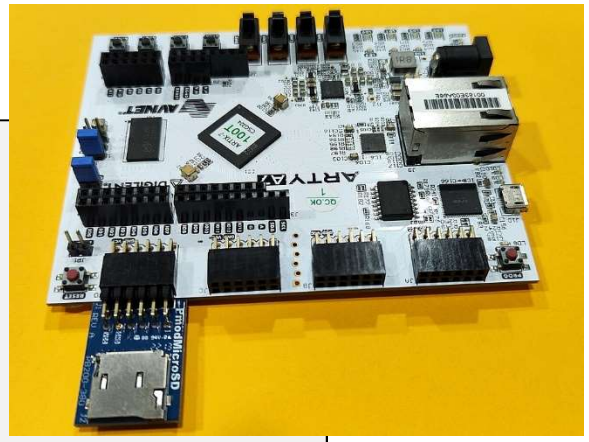
Example: AXI Quad SPI Controller

- Double-click the SD_Card IP opens the parameter box:



Instantiation of the IP in Aquila

- Instantiation of the SPI Controller:



```
// ----------------------------------
//  SPI controller
// ----------------------------------
//  This controller connects to the PMOD microSD module in
//  the JD connector of the Arty A7-100T.
//
axi_quad_spi_0 SD_Card_Controller(

    // Interface ports to the Aquila SoC.
    .s_axi_aclk(clk),
    .s_axi_aresetn(~rst),
    .s_axi_awaddr(axi_awaddr),
    .s_axi_awvalid(axi_awvalid),    // master signals write addr/ctrl valid.
    .s_axi_awready(axi_awready),    // slave ready to fetch write address.
    .s_axi_wdata(axi_wdata),        // write data to the slave.
    .s_axi_wstrb(axi_wstrb),        // byte select signal for write operation.
    .s_axi_wvalid(axi_wvalid),      // master signals write data is valid.
    .s_axi_wready(axi_wready),      // slave ready to accept the write data.
    .s_axi_araddr(axi_araddr),
    .s_axi_arready(axi_arready),    // slave ready to fetch read address.
    .s_axi_arvalid(axi_arvalid),    // master signals read addr/ctrl valid.
    .s_axi_bready(axi_bready),      // master is ready to accept the response.
    .s_axi_bresp(axi_bresp),        // response code from the slave.
    .s_axi_bvalid(axi_bvalid),      // slave has sent the respond signal.
    .s_axi_rdata(axi_rdata),        // read data from the slave.
    .s_axi_rready(axi_rready),      // master is ready to accept the read data.
    .s_axi_rresp(axi_rresp),        // slave sent read response.
    .s_axi_rvalid(axi_rvalid),      // slave signals read data ready.

    // Interface ports to the SD Card.
    .ext_spi_clk(clk),
    .io0_o(spi_mosi),
    .io1_i(spi_miso),
    .sck_o(spi_sck),
    .ss_o(spi_ss)
);
```

The Aquila Device Interface

- The Aquila core uses a simple memory-mapped I/O interface to talk to the external devices
  - A bridge is required for Aquila to talk to the AXI SPI controller

```
aquila_top Aquila_SoC
(
    .clk_i(clk), .rst_i(rst), .base_addr_i(32'b0),

    // External instruction memory ports.
    .M_IMEM_strobe_o(IMEM_strobe),
            . . .
    .M_IMEM_data_i(IMEM_data),

    // External data memory ports.
    .M_DMEM_strobe_o(DMEM_strobe),
            . . .
    .M_DMEM_data_i(DMEM_rd_data),

    // I/O device ports.
    .M_DEVICE_strobe_o(dev_strobe),     // Issue read/write requests.
    .M_DEVICE_addr_o(dev_addr),         // Target device address.
    .M_DEVICE_rw_o(dev_we),             // Read or write?
    .M_DEVICE_byte_enable_o(dev_be),    // Byte-select signal.
    .M_DEVICE_data_o(dev_din),          // Data input to the device.
    .M_DEVICE_data_ready_i(dev_ready),  // Is device ready?
    .M_DEVICE_data_i(dev_dout)          // Data output from the device.
);
```

Bridging the IP Interface

- Most IPs in the Xilinx IP Catalog use the AXI bus interfaces to communicate with other IPs:
  - AXI Full: enable both burst and single-beat data transfer
  - AXI Lite: enable single-beat data transfer
  - AXI Stream: enable burst-only data transfer

- We must convert the Aquila interface bus signals to the AXI bus signals for IP integration
  - The module `core2axi_if.v` is used for Aquila to connect to any IP that supports AXI Lite bus interface.

Running the CNN_OCR Program

1. First, copy the data files to the SD card
   → make sure the SD card is formatted as a FAT32 disk
2. Insert the SD card into the SD card pmod module
3. Insert the pmod module into the socket JD on Arty
4. Then, build `cnn_ocr.elf` by typing "`make`"
5. Synthesize the Aquila SoC and configure the FPGA
6. Load and run `cnn_ocr.elf`

Output of the CNN_OCR Program

- CNN_OCR make inferences on a sequence of images:

```
=========================================================================
Copyright (c) 2019-2025, EISL@NYCU, Hsinchu, Taiwan.
The Aquila SoC is ready.
Waiting for an ELF file to be sent from the UART ...

Program entry point at 0x80002958, size = 0xB90C.
-------------------------------------------------------------------------

(1) Reading the test images, labels, and neural weights.
It took 981 msec to read files from the SD card.

(2) Perform the hand-written digits recognition test.
Here, we use a 6-layer CNN neural network model.
Begin computing ... tested 100 images. The accuracy is 98.00%

It took 7915 msec to perform the test.

-------------------------------------------------------------------------
Program exit with a status code 0
Press <reset> on the FPGA board to reboot the cpu ...
```

Key Hotspot of the Program

- There are two functions that are good candidates of hardware acceleration:
  1. `conv_3d()` in `convolution_layer.h`
  2. `fully_connected_layer_forward_propagation()` in `fully_connected_layer.h`
- For the HW:
  - The second function computes 1D dot-product is much easier to accelerate, but give less speedup
  - The first function, which computes 3D dot-product, is the most time-consuming hotspot, but harder to turn into hardware
- Both circuits can use the 1D dot-product IP from Xilinx

Floating-Point IP in Xilinx IP Catalog

aquila_mpd - [R:/temp/aquila_dram/aquila_mpd/aquila_mpd.xpr] - Vivado 2024.1

File Edit Flow Tools Reports Window Layout View Help Quick Access write_bitstream Complete Default Layout

Flow Navigator

PROJECT MANAGER
- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR
- Create Block Design

SIMULATION
- Run Simulation

RTL ANALYSIS
- Run Linter
- Open Elaborated Design

SYNTHESIS
- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION
- Run Implementation

Click this to show the IP catalog!

PROJECT MANAGER - aquila_mpd

Sources

- Verilog Header (1)
- soc_top (soc_top.v) (8)
  - Clock_Generator : clk_wiz_0 (clk_wiz_0.xci)
  - Aquila_SoC : aquila_top (aquila_top.v) (6)
  - UART : uart (uart.v)
  - Core2AXI_0 : core2axi_if (core2axi_if.v)
  - SD_Card_Controller : axi_quad_spi_0 (axi_quad_spi_0.xci)
  - synchronizer : cdc_sync (cdc_sync.v) (10)
  - Memory_Arbiter : mem_arbiter (mem_arbiter.v)
  - MIG : mig_7series_0 (mig_7series_0.xci)

Hierarchy | IP Sources | Libraries | Compile Order

IP Properties

Floating-point

Version: 7.1 (Rev. 18)

Project Summary | IP Catalog

Cores | Interfaces

Search:

Name	AXI4	Stat
Math Functions		
Adders & Subtracters		
Conversions		
CORDIC		
Dividers		
Floating Point		
Floating-point	AXI4-Stream	Pro

We want to insert this floating-point IP that can compute inner-product.

bus interface of the IP

Details

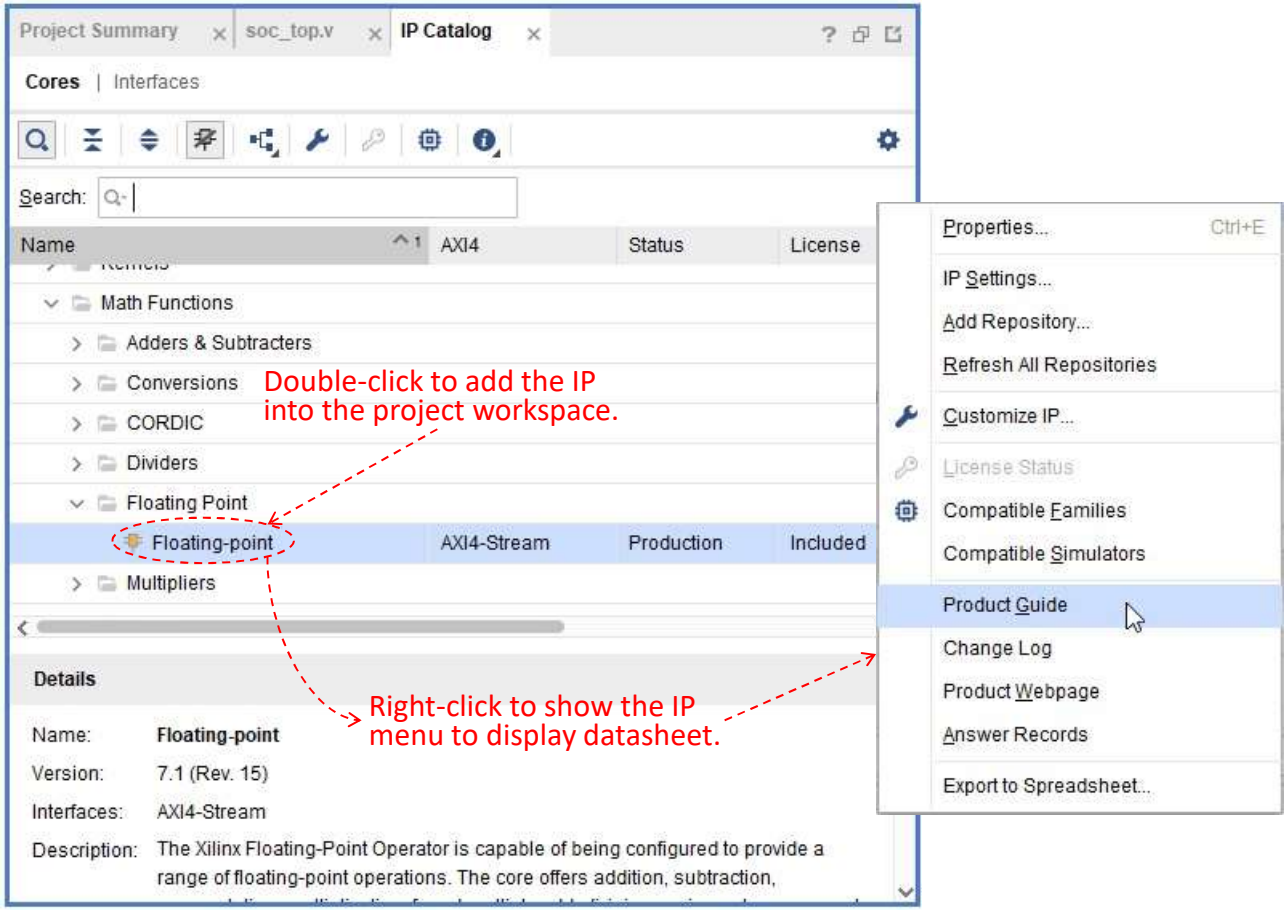
Name: Floating-point
Version: 7.1 (Rev. 18)
Interfaces: AXI4-Stream

Tcl Console | Messages | Log | Reports | Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology
synth_1 (active)	constrs_1	synth_design Complete!									
impl_1	constrs_1	write_bitstream Complete!	1.274	0.000	0.009	0.000	10.908	0.000	0.914	0	22 Warn

How to Add an IP into Aquila SoC

- Simply double-clicking the IP to add it to the workspace:

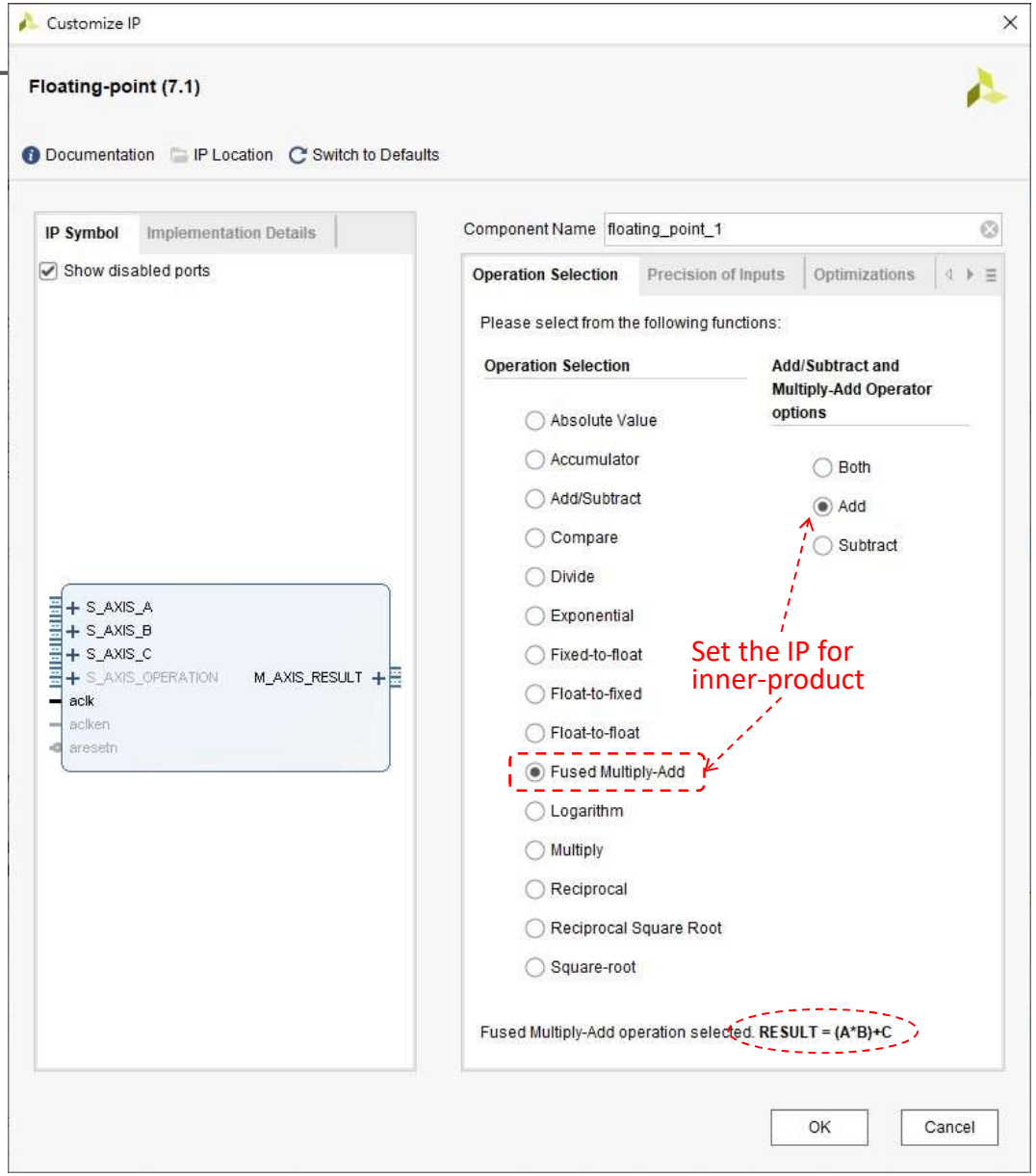


Configure the IP

- Double-clicking the IP, a pop-up dialog will show
  - Set the IP to do Fused Multiply-Add
- Default parameters:
  - Blocking mode
  - 32-bit precision
  - Latency: 17 cycles

Can be reduced to 2 without timing error on Arty. However, the latency is not an issue for this HW.

The bottleneck is somewhere else.



Inserting the IP from the TCL Script

- For HW submission, you should modify the TCL script to insert the floating-point IP into the workspace
  - To insert a fused multiply-add floating-point IP into the workspace, you can add the following code to `build.tcl`:

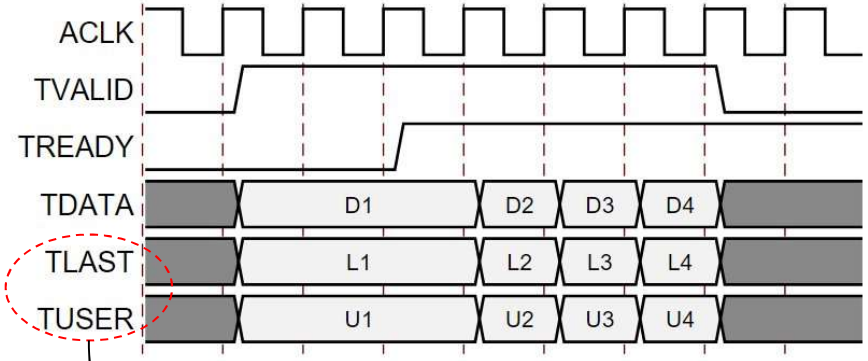
```
# Adding an Floating-Point IP
create_ip -name floating_point -vendor xilinx.com -library ip -module_name
floating_point_0
set_property -dict [list \
   CONFIG.A_Precision_Type {Single} \
   CONFIG.OPERATION_TYPE {FMA} \
   CONFIG.ADD_SUB_VALUE {Add} \
   CONFIG.Flow_Control {NonBlocking}] [get_ips floating_point_0]
generate_target all [get_files
${proj_name}/${proj_name}.srcs/sources_1/ip/floating_point_0/floating_point_0.xci]
```

  - The parameter options can be listed using the following TCL command in the TCL console window of Vivado:

```
report_property [get_ips floating_point_0]
```

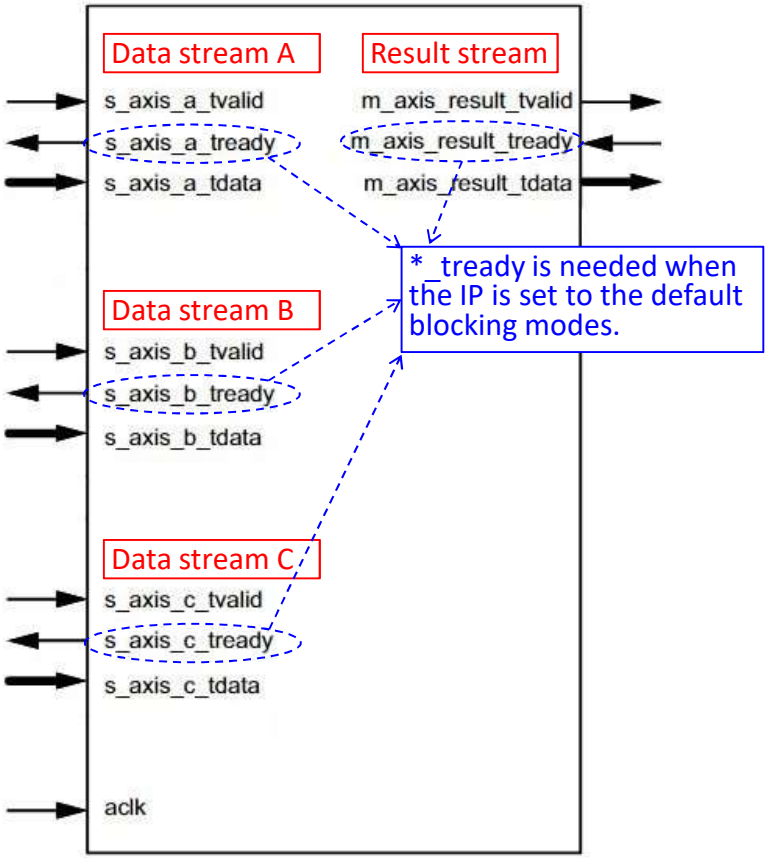
AXI Floating-Point IP Interface

- The floating-point IP is designed to handle a sequence of floating-point operations
- AXI stream bus is used for I/O ports:



TLAST and TUSER are optional for the floating-point IP.

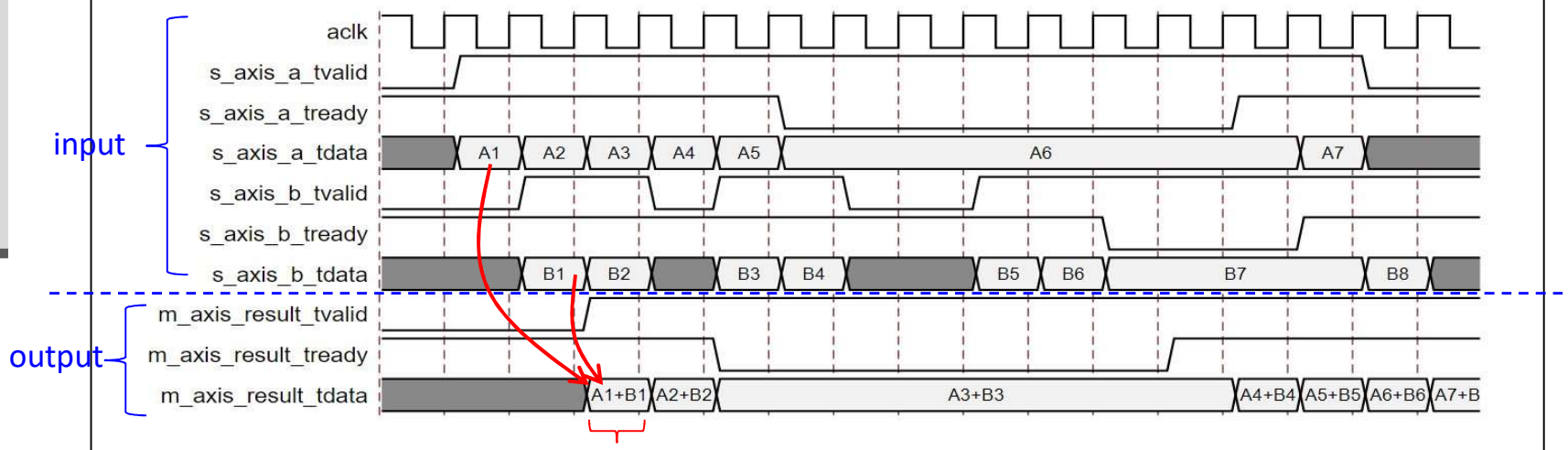
TLAST can be used to reset the accumulator when the IP is set to the "fused multiply-add" operation type, which is nice but not necessary.



Blocking vs Non-Blocking

- Two operation modes: blocking & non-blocking
  - Blocking mode requires controlling the extra ‘tready’ feedbacks
  - Non-blocking mode has no ‘tready’ feedbacks, assumes data will be fetched at the next clock rising edge after ‘tvalid’ is high!

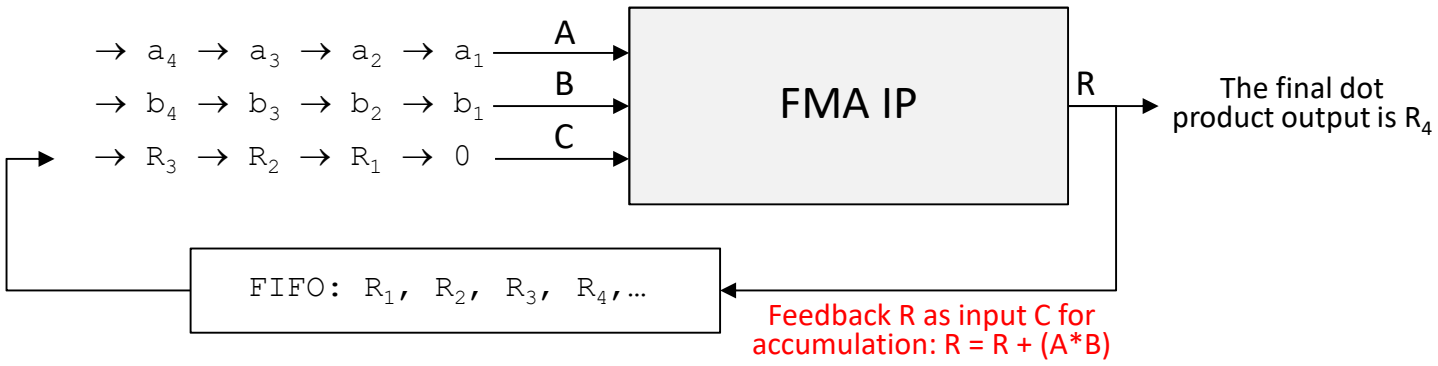
Timing diagram for blocking mode for floating-point “adder”



The two inputs A & B do not have to enter the IP at the same clock cycle in either mode!

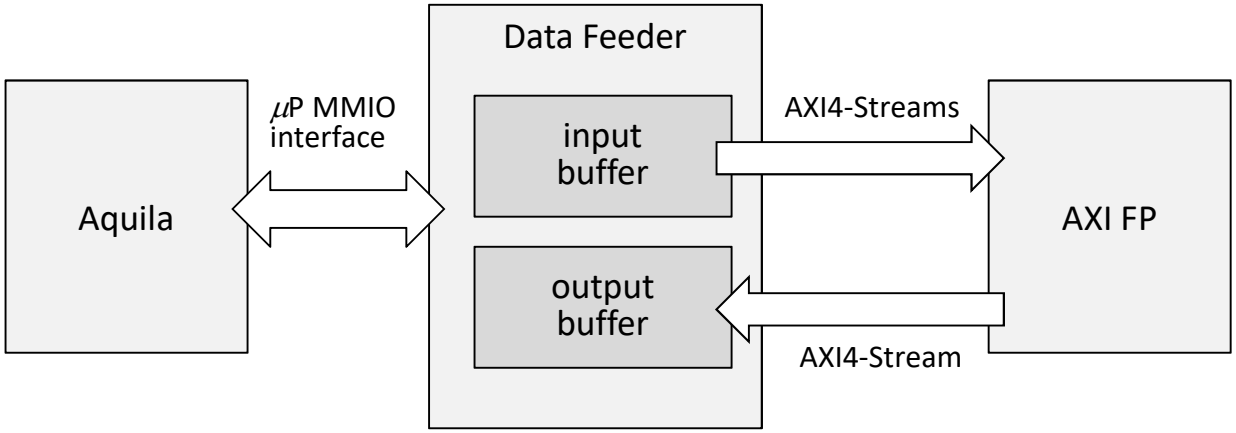
Blocking FMA IP Usage Example

- For dot-product, set the IP to fused multiply-add (FMA)
- For blocking mode, you must control 'tvalid' and 'tready' to align the input data
  - Assume that the vectors are 1×4, and the output R has a pipeline delay of 3 cycles (default delay is 17 cycles):



Interface between Aquila and HW FP

- You can design a data feeder module between the Aquila and the AXI FP:
  - On the Aquila-side, use `memcpy()` to copy the vectors into the input buffer, and read the final value from the output buffer
  - One the AXI FP side, use AXI4-stream bus to transmit the data
  - A good example of the μP MMIO interface is in `clint.v`



Examples of `conv_3d()` HW Interface

- The SW code shall be replaced by HW trigger code:

```
void conv_3d(uint64_t o, convolutional_layer *entry, float_t *pa)
{
    Code to setup pointers & indices!
    for (uint64_t inc = 0; inc < in_.depth_; inc++) {
        Code to setup pointers & indices!
        for (uint64_t y = 0; y < out_.height_; y++) {
            for (uint64_t x = 0; x < out_.width_; x++) {
                float_t *ppw = pw, sum = (float_t) 0;
                uint64_t wx = 0, widx = 0;
                for (uint64_t wyx = 0; wyx < weight_.height_ * weight_.width_; wyx++) {
                    sum += *ppw++ * ppi[widx];
                    wx++, widx++;
                    if (wx == weight_.width_)  {
                        wx = 0;
                        widx += const1;
                    }
                }
                pa[idx++] += sum;
                ppi += w_stride_;
            }
            ppi += const2;
        }
    }
}
```

Read soc_top.v, line 251 ~ 260.

```
volatile int *trigger_hw = (int *) 0xC4000000;

*trigger_hw = 1;
while (*trigger_hw) /* busy waiting */;
```

On the circuit side, please read clint.v to see how MMIO interface can be implemented!

Some Shortcuts to This HW

- To reduce the memory overhead, you can
  - Store the entire weight data into on-chip BRAM at the beginning of the program
    - This is not feasible for large CNN models, but ours is small
  - Feature maps of the previous layer can also fit into the BRAM
- In the CNN model, the paddings are 0 and the strides are 1, these constant can be used to simplify the functions to be made into accelerators

The Grading of This HW

- The grading is based on several factors:
  - Quality of the report
  - Performance
  - Whether FP IP is used
  - Whether weights/feature maps are stored in BRAM (not good)
  - Scalability of the design (to a larger AI model)

- Note that Using INT8 to replace all the computations will probably gives you the best performance
- Since the FP IP is much harder to use, you get extra points if you use FP IP and get good performance, say, at least two times faster than Aquila+FPU

Comments on the Homework

- For the accelerated system, one of the bottlenecks is in the feeding of data streams to the accelerator
  - You should measure the time spent on data feeding, and the time spent for computations, respectively
- The key point of this homework is to learn how to integrate an AXI accelerator to speed up computations
- Although no demo, you still need to upload your code to E3 so TAs can verify your results